

Boundary conditions in ASGarD

How a DG weak form becomes matrix blocks, how the flags select which blocks exist,
and how that composes across chains, terms and dimensions

Abstract. ASGarD has no boundary-condition module. Boundary conditions are an emergent property of *which matrix blocks get written during assembly*, and `boundary_type` is a switch selecting among three treatments of the two unpaired edge blocks at the ends of the domain. This document derives that architecture from the weak form, so that a reader can design boundary conditions for *new* terms rather than pattern-match existing ones. The spine: a `term_id` chain `{div, grad}` is a genuine matrix product $A = A_{\text{div}}A_{\text{grad}}$, formed at build time, with the LDG auxiliary variable eliminated algebraically. The `div`'s flag sets the numerical *flux* at a wall; the `grad`'s flag sets the numerical *trace*. A second-order operator needs exactly one condition per wall, so at each wall exactly one of the two links must impose its condition there. Everything else — the sum rule across independent terms, the emergent Robin condition, the meaning of a Robin coefficient, and the four ways to get it wrong — follows. Claims are tagged [INFERENCE] where inferred but untested and [UNRESOLVED] where the sources did not settle them; measurements are summarised in §10 and recorded in full, with commands, in `03_experiments.md` beside this file.

Source of truth. Line numbers refer to ASGarD 0.9.1 at `~/venvs/asgardpy/`; `include/` and `src/` are byte-identical apart from `assert↔expect`. The copy vendored in FENRIS is slightly older (`.cpp` lines shift by 10–20). FENRIS solvers alias `bc = asgard::boundary_type` and `flux = asgard::flux_type`; both spellings name the same enums.

1 Reference card

The one sign convention. `term_div(c)` assembles the DG weak form of $+\partial_x(cf)$ into a matrix A , and terms are applied with $\alpha = -1$.¹ **Throughout, F is the flux in the realised equation:**

$$F := cf, \quad \partial_t f = -\partial_x F, \quad dN/dt = F(a) - F(b). \quad (1)$$

Every weak form below describes A , *not* $\partial_t f$; the $\alpha = -1$ is paid once, here. If your PDE reads $\partial_t f = \partial_x G$ then $F = -G$ and you pass $c = -G/f$. Cross-checked against `sinwav.cpp`, which solves a right-moving wave with `term_div(+1, upwind, boundary_type::left)` plus a `left_boundary_flux` — correct only if `term_div(+1)` contributes $-\partial_x f$.

you write	assembly effect	physical condition
<code>term_div(c, none)</code>	write $\mp(c/h)$ to <code>to.*</code> into the boundary cell's diagonal	free / outflow: $\hat{F} = c f^{\text{int}}$ at the wall
<code>term_div(c, bothsides)</code>	write <i>nothing</i>	sealed: $\hat{F} = 0$, exactly, at both walls
<code>term_div(c, left)</code>	nothing at left, free block at right	$\hat{F} = 0$ at the left wall only
<code>term_div(c, periodic)</code>	interior formula into the wrap blocks	wall becomes an ordinary interior edge
<code>term_grad(c, none)</code>	flag flips to <code>bothsides</code> , nothing written	trace of f free
<code>term_grad(c, bothsides)</code>	flag flips to <code>none</code> , block written	trace $\hat{f} = 0$ (Dirichlet) at both walls
<code>term_grad(c, right)</code>	block written at the right wall	$\hat{f} = 0$ at the right wall
<code>term_penalty(p, bothsides)</code>	<i>add</i> $+(p/h)$ to <code>to.*</code>	Nitsche pin $f = 0$
<code>set_right_robin(r)</code> (chains only)	<i>add</i> $+(r/h)$ to <code>to_right</code>	wall flux $\hat{F} = rf$; needs $r \geq 0$
<code>+= left_boundary_flux(g)</code>	source vector $-cg \ell_L / \sqrt{h}$	inhomogeneous prescribed wall flux

The chain identity. A chain is the matrix product $A = A_{\text{div}}A_{\text{grad}}$, and in a dimension with mass weight m it realises

$$m \partial_t f = -\partial_x \left(\frac{c_d c_g}{m} \partial_x f \right) \implies \text{for physical } m \partial_t f = \partial_x (m D \partial_x f) : \quad \boxed{c_d c_g = -m^2 D}, \quad (2)$$

so c_d and c_g must have **opposite signs** (§4.2).

The flag rule. Read each flag as the *set of walls it names*: `none` $\mapsto \emptyset$, `left` $\mapsto \{L\}$, `right` $\mapsto \{R\}$, `bothsides` $\mapsto \{L, R\}$. The `div` names walls where it seals its flux; the `grad` names walls where it pins f . Then: **at each wall**,

¹`asgard_discretization.cpp:556`, and `...:630` for the implicit path.

exactly one of the two links must name that wall. Equivalently $\varphi_g = \text{flip}(\varphi_d)$ — the parity mnemonic, since `flip` is exactly set complement.

user flags	$\hat{q}$ (flux)	$\hat{f}$ (value)	realised
<code>div bothsides, grad none</code> (default)	0 both	interior	Neumann both walls, $\dot{N} = 0$
<code>div none, grad bothsides</code>	interior	0 both	Dirichlet $f = 0$ both walls
<code>div left, grad right</code>	0 at L	0 at R	Neumann at L , Dirichlet at R
<code>div bothsides, grad bothsides</code>	0	0	over-determined, diverges
<code>div none, grad none</code>	interior	interior	under-determined, asymmetric

The sum rule. Sealing a *set* S of terms enforces *one* condition per wall, $\sum_{k \in S} F_k \rightarrow 0$ — a *signed* sum evaluated *pointwise at the wall*. Sealing drag with diffusion therefore yields an *emergent Robin* $f'/f = -A/D$: the physics of an impermeable wall, and *not* $f' = 0$ (§5).

Five of the eleven pitfalls in §9. (1) Nothing validates flag combinations; every pairing compiles and runs. (2) Mass conservation does *not* catch the over-determined pair — it conserves mass while f diverges to 1.4×10^6 ; check $\max |f|$. (3) `flux_type::none` is silently `central`. (4) Coefficients are sampled *at* the endpoints, so flags are inert wherever the Jacobian vanishes there (v^2 at $v = 0$, $\sin \theta$ at $\theta = 0, \pi$). (5) Same-sign c_d, c_g is an *interior* instability, not a wall problem.

2 From the weak form to matrix blocks

The broken space. Partition $\Omega = [a, b]$ into cells $I_j = [x_{j-1/2}, x_{j+1/2}]$ of width h and let $V_h = \{v : v|_{I_j} \in P^k(I_j)\}$ with **no continuity between cells**. That single fact is the origin of everything below: a $w \in V_h$ has two values at every interior edge, w^- and w^+ , with $\llbracket w \rrbracket = w^+ - w^-$ and $\{\{w\}\} = \frac{1}{2}(w^+ + w^-)$. **At the two domain edges only one trace exists; the other would come from outside the domain. That asymmetry is the entire boundary-condition problem.**

Multiplying $\partial_x F$ by a test function $v \in P^k(I_j)$ and integrating over one cell — the only integration by parts in the problem — gives a *volume* term and a *surface bracket*. Since v lives on I_j alone, nothing about the neighbours enters except through the endpoints:

$$(Af)|_{I_j, v} = - \int_{I_j} v' F dx + v_{j+1/2}^- \hat{F}_{j+1/2} - v_{j-1/2}^+ \hat{F}_{j-1/2}, \quad F = c f_h. \quad (3)$$

On V_h the bracket is ambiguous, so DG replaces F at each edge by a **numerical flux** $\hat{F}$: one number per edge, shared by both neighbours. Sharing is not a convenience — it *is* the discrete statement of local conservation. ASGarD's edge blocks give the standard upwind flux

$$\hat{F}_{j+1/2} = c \{\{f\}\} - \frac{s|c|}{2} \llbracket f \rrbracket = \begin{cases} cf^- & c > 0, s = +1 \\ cf^+ & c < 0, s = +1 \end{cases} \quad s = \text{static_cast<int>}(flux) \in \{+1, 0, -1\}. \quad (4)$$

Nothing below depends on the degree k , the quadrature order, or on whether the basis is nodal Legendre or hierarchical wavelet — the wavelet transform is an orthogonal change of basis on the *same* V_h .

Where $1/h$ and $2/h$ come from. Legendre polynomials on $\xi \in [-1, 1]$ are rescaled so that $\frac{1}{2} \int P_i P_j d\xi = \delta_{ij}$;² the physical basis is $\varphi_j(x) = P_j(\xi)/\sqrt{h}$, making the cell mass matrix the identity (which is why nothing inverts a mass matrix unless you asked for one). An edge trace product is $\varphi_r(\pm 1)\varphi_c(\pm 1) = P_r P_c / h$ — one $h^{-1/2}$ per basis factor — while $d\varphi_c/dx = (2/h)P'_c/\sqrt{h}$ with $\int_{\text{cell}} dx = (h/2) \int d\xi$ and the quadrature array already carrying the $\frac{1}{2}$. Those are exactly the two code constants, `escale` = $1/dx$ and `vscale` = $2/dx$.³ No $\sqrt{h}$ survives: the operator is dimensionally $1/\text{length}$, as ∂_x should be.

The four edge blocks. Blocks are column-major with row = test, column = trial. With $\ell_L = P(-1)$, $\ell_R = P(+1)$ they are rank-one outer products,⁴ built once:

²`asgard_quadrature.cpp:58–66.`

³`asgard_coefficients_mats.hpp:187–188.`

⁴`asgard_transformations.cpp:55–58; gemm_outer_inc(n,x,y,A)` does `A[c*n+r] += x[r]*y[c]`, `asgard_small_mats.hpp:593–600.`

<code>to_left</code> = $\ell_L \otimes \ell_L$	<code>from_left</code> = $\ell_L \otimes \ell_R$	<code>from_right</code> = $\ell_R \otimes \ell_L$	<code>to_right</code> = $\ell_R \otimes \ell_R$
test trace at my LEFT edge $\times$	at my LEFT edge $\times$	at my RIGHT edge $\times$	at my RIGHT edge $\times$
my own value there	left neighbour's value	right neighbour's value	my own value there

`to_*` means the trace comes from *this* cell (diagonal block); `from_*` from the *neighbour* (off-diagonal). **These four objects are the entire vocabulary of boundary conditions in ASGarD: every boundary condition in the code is a decision about whether to write a multiple of `to_left` or `to_right` into the first or last diagonal block.**

The operator is block-tridiagonal, with `lower(0)` and `upper(N-1)` reserved as wrap blocks for periodicity.⁵ For an interior cell i , with $\ell = \frac{1}{2}c(x_{i-1/2})$, $r = \frac{1}{2}c(x_{i+1/2})$, the assembly writes⁶ `lower(i) +=  $\frac{1}{h}(-\ell - s|\ell|)$ from_left`, `diag(i) +=  $\frac{1}{h}(-\ell + s|\ell|)$ to_left +  $\frac{1}{h}(r + s|r|)$ to_right`, `upper(i) +=  $\frac{1}{h}(r - s|r|)$ from_right` — which is (3) with (4). For `upwind` and $c > 0$ this collapses to `lower =  $-(c/h)$ from_left`, `diag =  $+(c/h)$ to_right`: the trace is taken from the *left* at every edge, which with (1) is genuine upwinding.

2.1 Telescoping: only two blocks can break conservation

Set $v \equiv 1$ in (3) — legitimate for every $k \geq 0$, since constants lie in P^k on each cell. The volume term dies because $v' = 0$, and with $\partial_t f = -Af$ the cell mass $m_j = \int_{I_j} f_h$ obeys

$$dm_j/dt = \hat{F}_{j-1/2} - \hat{F}_{j+1/2}. \quad (5)$$

Each cell is an exact ledger. Sum over j . Every **interior** flux appears exactly twice — with $-$ from cell I_j and $+$ from cell I_{j+1} — and because it is *the same number* in both, the two cancel identically. Structurally: each interior edge contributes a *matched pair* of blocks, one `to_*` in cell i 's diagonal and one `from_*` in the neighbour's off-diagonal, and tested against φ_0 the pair cancels exactly. So

$$\boxed{d/dt \int_{\Omega} f_h = \hat{F}_a - \hat{F}_b} \quad (6)$$

which is (1) at the discrete level. Three readings:

- The cancellation is **algebraic, not asymptotic**. It does not improve with $h \rightarrow 0$ or $k \rightarrow \infty$; it is exact at every resolution, for any upwinding parameter s , for any coefficient $c(x)$ however rough, and for any quadrature rule — the volume term never entered (6) at all.
- Therefore **the interior discretisation cannot create or destroy mass**. It is incapable of it. Every unit of mass that appears or disappears in an ASGarD run passed through one of exactly two numbers.
- Consequently, *for the conservative content of a term*, a boundary condition is a rule for those two numbers, and there is nowhere else to put one. The qualifier is load-bearing: a second-order term has two *more* wall degrees of freedom, the traces $\hat{f}_a, \hat{f}_b$ of §4, and those do not enter (6) at all. That is exactly why the over-determined pairing conserves mass perfectly while diverging pointwise (P2).

Only the two outermost edge blocks are unpaired. When $N > 1$, cell 0's right edge and cell $N-1$'s left edge are ordinary interior edges written unconditionally; only the outermost two are governed by the boundary switch.⁷

3 What each boundary_type does at assembly

The whole treatment is three `switches` in one `#pragma omp single` block.⁸ At the left wall, for a `div`:

```

1 case boundary_type::none:
2 case boundary_type::right: // free on the left
3     smmat::axpy(nblock, -escale * (c at xleft), basis.to_left, coeff.diag(0)); break;
4 default:
5     // dirichlet flux, nothing to set
6     break;
7
8
9
```

⁵ `asgard_block_matrix.hpp`:317–320; slots always exist (`asgard_grid_1d.hpp`:316–327) and are zero otherwise.

⁶ `asgard_coefficients_mats.hpp`:239–249; the volume block by `apply_volume` at ...:210–222 as $-c \int \varphi'_r \varphi_c$.

⁷ `asgard_coefficients_mats.hpp`:321–333; `num_cells == 1` special-cased at ...:263–264, 321.

⁸ `asgard_coefficients_mats.hpp`:255–351; enum at `asgard_pde.hpp`:109–121. Interior cells are handled by the parallel loop above.

⁹ `asgard_coefficients_mats.hpp`:305–319; the right wall at ...:336–350 mirrors it, writing `+escale*(c at xright)*to_right` into `diag(rmost)`.

none — **free**. An unpaired block $\mp(c/h)\text{to_}\ast$ goes into the boundary cell’s own diagonal; tested against φ_0 it is $\hat{F} = c f^{\text{int}}$. The coefficient is the *full* c , not $c/2$, and $|c|$ never appears — upwinding is abandoned at the wall. **This is only well-posed on outflow**. When the characteristic points *in*, extrapolating posits a reservoir outside the domain whose density equals the solution’s own wall trace, and (6) acquires $\dot{N} \ni -c f^{\text{int}}(b)$: a source *proportional to the unknown*. That is an $O(1)$ modelling error, not a discretisation error, so refinement converges faster to the wrong answer. It is the ICRF_2D pathology — its drag has $c(x_{\text{max}}) < 0$, so $\dot{N}|_b > 0$: +63% manufactured particles by $t = 100$, worse under refinement. The code runs silently; [INFERENCE] that the absence of a guard is oversight rather than design.

bothsides — **sealed**. The **default**: arm, commented “*dirichlet flux, nothing to set*”. **No axpy executes**: the block is omitted, so $\hat{F} = 0$ identically — enforced by *absence*, hence exact to machine precision at every level and degree.

The “dirichlet flux” trap. The comment means Dirichlet *on the flux*, $\hat{F} = 0$ — which as a condition on f is a **Neumann** condition for a diffusive term and a no-outflow condition for an advective one. It is *not* $f = 0$. Confusing the two is the commonest way to mis-flag a term.

left/right are **bothsides** on one wall only; the flag names the wall where the flux *is* pinned. **periodic** writes the ordinary interior formula into the wrap blocks, so the domain edge becomes a genuine interior edge joining cell $N-1$ to cell 0 and conserves by the same pairwise cancellation. [UNRESOLVED]: nothing cross-checks flags *between* terms, so mixing **periodic** and **bothsides** in one dimension across two terms was not established.

Sampling, and inert flags. The wall values used above are c evaluated **exactly at the domain endpoint**.¹⁰ So **wherever the flux coefficient vanishes at the wall, none and bothsides assemble the same matrix** and the seal is a no-op. In FENRIS this is routine: $c \propto v^2$ makes the $v = 0$ wall inert, and $c \propto \sin \theta$ on $[0, \pi]$ makes *both* θ walls inert. Never conclude from “that wall is fine” that the flags are right — they may be inert. Structural (from the sampling location), not separately measured.

3.1 grad: the flag is flipped, but you get the wall you named

Before assembly, a **grad**’s flag is flipped — **bothsides** $\leftrightarrow$ **none**, **left** $\leftrightarrow$ **right**, **periodic** untouched “since it is symmetric anyway”.¹¹ A div-style matrix $\tilde{A}$ is built with the *flipped* flag and then replaced by its negative transpose with off-diagonals swapped,¹² giving $G = -\tilde{A}^T$: the discrete $\partial_x^* = -\partial_x$. Since **to_left** is symmetric, a free block $-(c/h)\text{to_left}$ in $\tilde{A}$ becomes $+(c/h)\text{to_left}$ in G , so $(Gf)_r = c \int \varphi_r \partial_x f + c \varphi_r(x_0) f(x_0) + \dots$ — which against the strong-form DG discretisation of $c \partial_x f$ is exactly the wall correction with $\hat{f} = 0$. So:

For a **grad**, the **presence** of the boundary block pins the numerical trace of f at that wall to **zero** (**Dirichlet**); its **absence** takes the trace from the interior.

Because the flag was flipped, “block present” corresponds to the *user* writing **bothsides**. **The flip exists precisely so that bothsides means “impose my condition here” uniformly, whichever link you write it on.**

Which wall. The internal **right** $\rightarrow$ **left** flip is bookkeeping for the div-style assembler; the negative transpose puts the block back where you asked. Trace it: user **grad right** $\rightarrow$ flipped **left** $\rightarrow$ left switch seals (no block), right switch writes $+(c/h)\text{to_right}$ $\rightarrow$ after $-\tilde{A}^T$ the block sits in $\text{diag}(N-1)$. **So term_grad(c, right) pins f at the right wall**, confirmed by measurement: $\{\text{div bothsides, grad right}\}$ diverges at $f(2)$, not $f(0)$.

A second flip, at **term_1d** construction, turns a user-written **upwind** on a **grad** into a stored **downwind**.¹³ Both **term_div** and **term_grad** default to **flux_type::upwind** and **boundary_type::none** and take the two arguments in either order,¹⁴ so writing nothing about the flux gives **upwind** on both links and opposite *stored* fluxes — half of what the interior alternation needs; see §4.2 for the other half.

¹⁰asgard_coefficients_mats.hpp:164–182.

¹¹asgard_coefficients_mats.hpp:130–148.

¹²asgard_coefficients_mats.hpp:355–366, including a final swap of the periodic wrap pair.

¹³asgard_pde.hpp:719–724.

¹⁴asgard_pde.hpp:171–215.

3.2 penalty: bothsides adds instead of deletes

A penalty has **no volume integral**, and its interior blocks are a pure jump form $p[\![\varphi]\!][\![f]\!]$, so $f^\top P f = +p \sum_e [\![f]\!]^2 \geq 0$ and with $\alpha = -1$ the contribution is dissipative.¹⁵ Its boundary switch puts **bothsides** in the arm that **adds** $+(p/h)\text{to_left}$ and **none** in the **default**: that adds nothing — the exact opposite of the **div** switch.¹⁶ The inversion dissolves once you see *what* is added: $p\varphi_r(x_0)(f(x_0) - 0)$, a one-sided jump against a *zero exterior state*, i.e. Nitsche-style weak enforcement of $f = 0$. **For div/grad the flag names a flux to delete; for penalty it names a value to pin.** The penalty flag is not flipped. `set_penalty(p)` on a link inherits the parent's flag, on a chain the *last* link's,¹⁷ which makes both canonical pairings self-consistent — but `div(bothsides)` plus a penalty deletes the flux block (Neumann) and adds a Dirichlet pin, two conditions on one wall. **[INFERENCE]**: over-determined like §4.3(iii); read from the dispatch, not tested.

3.3 Inhomogeneous wall data is a source vector

`left_boundary_flux/right_boundary_flux` build $-cg\ell_L/\sqrt{h}$ and add it to the right-hand side with the negated weight¹⁸ — precisely the omitted boundary block with the unknown trace replaced by g . So `boundary_type::left + left_boundary_flux(g)` prescribes $\hat{F}_a = g$, and sealing alone is the homogeneous case (`sinwav.cpp` is the worked example). The supplied function must be constant in the *flux* dimension but **may vary in the others**.¹⁹ you cannot prescribe a profile *across* the wall, but you can prescribe one *along* it, and this is the only mechanism in the API for wall data depending on another coordinate. For a chain, `chain_level(d)` selects the link and the vector is multiplied through the outer links.²⁰

4 Chains: the matrix product and the per-wall rule

4.1 Two wall traces, not one

For $\partial_t f = \partial_x(D\partial_x f)$ the naive move — integrate by parts once and write $[vD\hat{f}']$ — fails structurally: $D\partial_x f_h$ is the derivative of a broken function, so across an edge it contains a Dirac mass the cellwise derivative discards. One cannot integrate by parts twice on a space that is not H^1 . LDG refuses to, splitting into a first-order system with an auxiliary q and applying §2's machinery *twice*:

$$\begin{aligned} \int_{I_j} w q_h &= - \int_{I_j} c_g w' f_h + c_g w_{j+1/2}^- \hat{f}_{j+1/2} - c_g w_{j-1/2}^+ \hat{f}_{j-1/2}, & (\text{grad}) \\ (Af)|_{I_j,v} &= - \int_{I_j} v' c_d q_h + v_{j+1/2}^- \hat{q}_{j+1/2} - v_{j-1/2}^+ \hat{q}_{j-1/2}. & (\text{div}) \end{aligned}$$

Two integrations by parts have produced **two independent numerical traces**: $\hat{f}$ appears *only* in the **grad** equation, $\hat{q}$ *only* in the **div** equation. **(grad)** has no place to put a condition on the flux; **(div)** has no place to put one on the value.

the **div** flag sets $\hat{q} \Leftrightarrow$ a condition on the *flux*; the **grad** flag sets $\hat{f} \Leftrightarrow$ a condition on the *value*.

(7)

They are not two chances to say the same thing; they are two *different* degrees of freedom at each wall. Note $\alpha = -1$ has *not* been applied here: **(grad)**–**(div)** describe A .

4.2 The product, and what a chain realises

`rebuild_chain [sic]` builds the *last* link first and multiplies leftwards, so the stored operator is²¹ $A = A_{\text{div}} A_{\text{grad}}$. $q = A_{\text{grad}} f$ is a **virtual vector**: never assembled, never stored, never given its own boundary condition. It exists only as the middle of a product collapsed at build time. *That is why the two flags interact.*

¹⁵`asgard_coefficients_mats.hpp`:212, 191, 120–121, 231–235; algebraically $\text{div}_{\text{upwind}}(c) = \text{div}_{\text{central}}(c) + (|c|/2) \cdot \text{penalty}$.

¹⁶`asgard_coefficients_mats.hpp`:268–279.

¹⁷`asgard_term_build.cpp`:838–841, 851–854, 1095–1107.

¹⁸`asgard_term_build.cpp`:890–931; negation at `asgard_term_sources.cpp`:222, 237–239.

¹⁹`asgard_pde.hpp`:1288–1289.

²⁰`asgard_term_build.cpp`:881–887. *Outermost* throughout this document means the first entry in the chain list, i.e. the left factor of $A_{\text{div}} A_{\text{grad}}$; `chain_level` defaults to -1 , which attaches there.

²¹`asgard_term_build.cpp`:988–1092; design stated at `asgard_pde.hpp`:384–386.

Since $A_{\text{div}} \approx \partial_x(c_d \cdot)$, $A_{\text{grad}} \approx c_g \partial_x$, and M^{-1} is applied *per link* (§7), the realised PDE in a dimension with mass weight m is

$$\boxed{m \partial_t f = -\partial_x \left(\frac{c_d c_g}{m} \partial_x f \right) \quad \implies \quad \text{for } m \partial_t f = \partial_x (m D \partial_x f) : \quad c_d c_g = -m^2 D.} \quad (8)$$

With no Jacobian, $c_d c_g = -D$. Two production instances: **ICRF_1D**, $m = v^2$, $c_d = -\zeta v^2$, $c_g = +v^2$, product $-\zeta v^4 = -m^2 \zeta$, so $D = \zeta$;²² and **LHCD_2D** radial, $m = x^2$, $c_d = -0.25$, $c_g = x$, product $-0.25x = -x^4 \cdot \frac{1}{4x^3}$, so $D = 1/(4x^3)$ — exactly what that solver’s own comment states its collisional radial diffusivity to be.²³

Only the *product* fixes the PDE; the split between the links is free. The FENRIS idiom puts the Jacobian on *both* links so the per-link M^{-1} cancels one factor in each, leaving $M^{-1} A_{\text{grad}} \approx (1/v^2)(v^2 \partial_v) = \partial_v$. What the **div** flag seals is $c_d q$: the mass-weighted flux, whose wall value controls $d/dt \int m f$.

The sign is not free, and it is the same requirement as interior stability. (8) with $D > 0$ forces c_d, c_g to have **opposite signs**. That is not a coincidence. The LDG interior alternation — the **div** taking its trace from one side of each edge and the **grad** from the other — needs the two upwind splits $c \pm |c|$ to select opposite sides, and the constructor’s **upwind**→**downwind** flip supplies only half of that; the other half is the sign of each coefficient. Worked for constant c with **upwind** declared on both links: a **div** with $c_d > 0$ takes $\hat{q} = q^-$ and a **grad** with $c_g > 0$ takes $\hat{f} = f^-$ — *both from the left*, no alternation. Making c_d negative moves the **div**’s trace to q^+ and restores it. So:

c_d and c_g must have opposite signs. Getting the physical diffusivity sign right and getting the LDG alternation right are the same condition — and nothing checks either.

This is why P6’s blow-up is possible and why it is *interior*, not a wall effect. **check_chain** enforces only “at most two non-central fluxes, never a central mixed with a side flux”; it does not verify that the two side fluxes point opposite ways and cannot know the sign of a variable coefficient.²⁴ **[INFERENCE]**: worse, **gemm_block_tri** computes only three bands (**asgard_block_matrix.hpp**:544–547), relying on the two factors being bidiagonal in *opposite* directions; otherwise the out-of-band entries are silently dropped and you get a consistent-looking but wrong operator. Read from source, not measured.

Chain hygiene: identity links are stripped, a single-non-identity chain collapses to that term, and chains of chains throw,²⁵ so **{div, identity, grad}** and **{div, grad}** build the same matrix. Volume and identity terms have no edge blocks at all.

4.3 The rule, stated per wall

Read each flag as the **set of walls it names** (**none** $\mapsto \emptyset$, **left** $\mapsto \{L\}$, **right** $\mapsto \{R\}$, **bothsides** $\mapsto \{L, R\}$). Let S_d be the walls where the **div** seals its flux ($\hat{q} = 0$) and S_g the walls where the **grad** pins the trace ($\hat{f} = 0$). A second-order operator on an interval needs exactly one condition per wall, so

$$\boxed{\text{at each wall, exactly one of the two links must name that wall: } S_g = \mathbb{C} S_d.} \quad (9)$$

Since **flip** is exactly set complement on this encoding, (9) is literally $\varphi_g = \text{flip}(\varphi_d)$, the parity mnemonic — easier to remember, but it does not tell you *which* of Neumann and Dirichlet you bought. Note (9) does *not* say “exactly one link carries **bothsides**”: **{div left, grad right}** carries none and is perfectly legal. **periodic** sits outside the framing ($\text{flip}(\text{periodic}) = \text{periodic}$) and must be matched **periodic**↔**periodic**; see the **[UNRESOLVED]** in §3.

The five cases are tabulated in §1; what that table cannot say is what each one does. **(i) Sealed pair** ($S_d = \{L, R\}$, $S_g = \emptyset$): the **div**’s block is deleted, so $\dot{N} = 0$ *exactly*, while the **grad** still sees the true interior trace, so q_h is computed correctly right up to the wall. Any constant is a steady state; all four FENRIS solvers use this. **(ii) Open pair**: mass leaves, as an absorbing wall must. **(iii) Both name a wall** asserts $f = 0$ and $\partial_x f = 0$ at one point; measured, divergence to 1.4×10^6 at level 7 growing as h^{-3} , *with mass conserved* (P2). **(iv) Neither names a wall**: the outcome is decided by the upwinding convention, which for a parabolic operator has no

²²FENRIS/ICRF_1D/src/ICRF_1D.cpp:257, 276–287.

²³FENRIS/LHCD_2D/src/LHCD_2D.cpp:161, 175–185.

²⁴asgard_pde.hpp:730–745.

²⁵asgard_pde.hpp:524–561.

physical meaning — measured, the left wall converges to ≈ 0.192 (19% below the clean Neumann 0.237950) and the right to 0 at $O(h)$ and slightly *negative*, despite symmetric flags. **(v) Mixed one-sided** is legal and is the common design (§8.1); not measured, but it follows from (i), (ii) and locality.

Locality. The bracket in (3) is a sum of two *separate* endpoint terms, so the walls are algebraically independent and a bad flag at one cannot make the other diverge. Measured for `{div bothsides, grad right}`: $f(2)$ diverges as h^{-3} while $f(0)$ stays bounded and converges at $O(h)$. **But the good wall is not “exactly Neumann”**: it reads 0.201417 at level 7 (extrapolating to ≈ 0.2018) against the clean Neumann 0.237950 — 15% off, because the boundary spike redistributes the conserved mass. Only the *divergence* is local; the *error* is global.

Energy check. Summing the two weak forms with $v = f_h$, $w = q_h$ over cells gives the LDG energy identity, whose interior part cancels identically under the alternating flux and whose right-wall part is $B_b = \hat{f}_b \hat{q}_b - (f^- - \hat{f}_b)(q^- - \hat{q}_b)$ (full derivation in `02_weak_form.md` §7). Since nothing constrains the signs of f^- , q^- independently, $B_b \leq 0$ for all data requires killing both products: set *one* trace to zero and the *other* to the interior value — which is (9), derived rather than asserted. Both consistent pairings give $B_b = 0$ and hence $\frac{1}{2} d\|f_h\|^2/dt = -\|q_h\|^2 \leq 0$: unconditional dissipation at any h , any k . Both inconsistent pairings leave $\pm(fq)^-$, of no fixed sign. The left-wall mirror carries the opposite sign, which is why case (iv) is left–right asymmetric although its flags are symmetric — consistent with the measured mongrel (the argument predicts asymmetry; it does not by itself predict the measured net mass loss).

5 The sum rule across independent terms

Real kinetic equations have several flux channels, $F = \sum_k F_k$, each assembled as its own term with its own flag.

$$\boxed{\text{Sealing a set } S \text{ imposes the single weak condition } \sum_{k \in S} F_k \rightarrow 0 \text{ at the wall.}} \quad (10)$$

One condition per wall, not one per term. The assembly deletes each sealed term’s block *individually*, so one might expect $|S|$ conditions. It does not.

Derivation. The weak form is linear in the terms, so $\hat{F} = \sum_k \hat{F}_k$ and (6) becomes

$$dN/dt = \sum_{k \notin S} [\hat{F}_k(a) - \hat{F}_k(b)], \quad (11)$$

since sealed terms contribute nothing at the wall: **the leak rate is the sum of the unsealed fluxes.** For the non-trivial half, apply the ledger (5) to the last cell I_N , whose right edge *is* the wall: $dm_N/dt = \sum_k \hat{F}_k(x_{N-1/2}) - \sum_{k \notin S} \hat{F}_k(b)$. Note the asymmetry: at the interior edge $x_{N-1/2}$ there are no flags, so *all* terms contribute; at the wall only the unsealed ones do. Now $m_N = O(h)$, so $dm_N/dt = O(h)$, and letting $h \rightarrow 0$ (the trace is continuous to $O(h^{k+1})$ across the last cell) gives $\sum_k F_k(b) = \sum_{k \notin S} F_k(b)$, i.e. (10). **No steady-state assumption is needed**, but the rule is asymptotic in h : exact on the numerical fluxes, $O(h^2)$ on the post-processed pointwise f'/f . One equation — the individual sealed fluxes are *not* separately zero, only their sum is. Structurally the mechanism is trivial: all terms accumulate into one right-hand side,²⁶ each open term depositing its unpaired wall block and each sealed term depositing nothing. **There is no per-term boundary condition anywhere in the code — only one flux budget per wall, and the flags choose who is in it.**

Three regimes. With $A_S = \sum_{k \in S} a_k(x_w)$ and $D_S = \sum_{k \in S} D_k(x_w)$ at the wall, (10) reads $A_S f + D_S f' = 0$, so: $D_S > 0$ gives $f'/f = -A_S/D_S$ (**emergent Robin**); $D_S = 0$ with $A_S \neq 0$ gives $f = 0$ (**Dirichlet**); $D_S = A_S = 0$ gives no condition. The first is the important one: sealing drag together with diffusion gives $f'/f = -A/D$ — **a Robin condition that nobody wrote**. That is the correct physics: an impermeable wall in the presence of drag does not give $f' = 0$; it gives the drag–diffusion balance, the Maxwell–Boltzmann-like exponential foot a no-flux kinetic equilibrium must have. Sealing gets it for free; imposing $f' = 0$ by hand would be *wrong*. The second is easy to trigger by accident: sealing a set with no f' channel pins f , not f' (measured $f(2) = 1.7 \times 10^{-9}$ against an interior scale 4.9). The first also explains why sealing *one* diffusion channel silences the others: $A_S = 0$ forces $f'(x_w) = 0$, which zeroes the wall flux of *every* purely diffusive channel there, whatever its coefficient — measured, sealing $\{D_1, D_2\}$, $\{D_1\}$, $\{D_2\}$ gives leak rates 9.01295/9.01295/9.01294, identical to six figures, although the open diffusivity is 0, 0.5, 1.0.

Signed, and strictly local. These two properties distinguish (10) from every plausible alternative, and both were measured. With two drags of opposite sign, sealing only the negative one gives $A_S = -1$ and predicts a wall

²⁶`asgard_term_manager.cpp:103–120.`

slope of the *opposite* sign, $+0.66667$: measured to 2.8×10^{-7} . Tuning them to cancel gives $A_S = 0$ and a pure Neumann wall: measured $f' = 3.0 \times 10^{-13}$. No “first term wins”, “largest term wins” or “sum of magnitudes” rule produces either number. And replacing constant $D_2 = 0.5$ by $D_2(x) = 0.25x$ — a different function throughout the interior, changing the whole interior solution — gives the *same* wall slope to 5 digits, because it has the same value *at the wall*.

Structural versus asymptotic, and confusing them causes false alarms. *Conservation is structural*: with everything sealed, mass is conserved to 10^{-15} at every level, because the block is absent from the matrix. *The pointwise wall condition is asymptotic*: measured order 2.00–2.02 in h , so a diagnostic checking the wall slope must expect $\approx 2.5 \times 10^{-3}$ at level 4 and $\approx 1 \times 10^{-5}$ at level 8.

6 Robin conditions

The whole of `gen_robin_cmat` that touches the matrix is two `axpys`:²⁷ `-(ra/h)to_left` into cell 0 and `+(rb/h)to_right` into the last cell. These are **the same blocks, with the same signs and the same $1/h$** , that the `none` branches of §3 write — with r in place of c . So a Robin block re-installs exactly the free/outflow wall block with a user-chosen coefficient: $\hat{F}^R(a) = r_a f(a)$, $\hat{F}^R(b) = r_b f(b)$.

r is a flux coefficient, not a log-derivative. Because r sits where a flux sits, $[r] = [F]/[f] = \text{flux per unit } f$: for a velocity coordinate, an *effective velocity* (times whatever Jacobian the term carries). It is not f'/f , whose dimension is $1/\text{length}$. Passing A/D to `set_right_robin` is dimensionally wrong, and this is not hypothetical: ICRF_2D’s recorded history is that $r = A_{cv}/B_{cv} = 2x_{\max}$ — precisely the log-derivative — *overshoots by $\sim 14\times$ and the solve diverges*, while $r = A_{cv}$, the term’s own wall flux coefficient, agrees with plain `bothsides` to solver tolerance.²⁸

Equivalence to sealing. A free `div` contributes $\hat{F}_b = c(b)f^-$; adding a Robin gives $(c(b) + r_b)f^-$. Hence

$$\boxed{\text{free}(c) + \text{Robin}(r = -c(\text{wall})) \equiv \text{sealed}(c),} \quad (12)$$

an identity of *matrices*: both multiply the same rank-one block, and when $r_b = -c(b)$ exactly the two `axpys` cancel to 0 in IEEE arithmetic. Measured 2.1×10^{-13} in a toy where r was a hand-typed literal against a quadrature-evaluated $c(b)$, and *bit-identical* in ICRF_1D and LHCD_2D where the two are the same expression at the same point.

The sign condition. The right-wall block adds `+(rb/h)to_right` to A , so $f^T A f \ni r_b (f^-(b))^2$; with $\partial_t f = -A f$, $\frac{1}{2} d\|f\|^2/dt \ni -r_b (f^-(b))^2 + r_a (f^+(a))^2$, giving

$$\boxed{\text{dissipative iff } r \geq 0 \text{ at the } \textit{right} \text{ wall and } r \leq 0 \text{ at the } \textit{left}.} \quad (13)$$

The mass balance agrees: $\dot{N} \ni r_a f(a) - r_b f(b)$, so an absorbing right wall needs $r_b > 0$ and an absorbing left wall $r_a < 0$. A Robin of the wrong sign is an energy source of strength $|r|f^2$ — a positive feedback on the wall value, which is the free-flux-on-inflow pathology seen from the energy side. Sanity check against (12): sealing a term with $c(b) < 0$ is spelled $r_b = -c(b) > 0$, on the dissipative side, as it must be. Both production Robins were of that form: ICRF_1D’s $r = +v_{\max}^2 \eta(v_{\max})$ against $c = -v^2 \eta$, and LHCD_2D’s $r = +0.5$ against $c = -0.5$.

Restrictions, all in the API. `set_left_robin` / `set_right_robin` take a single scalar and `assert(is_chain())`:²⁹ r cannot vary in space or time and cannot attach to a bare `term_div`. It *is* mass-weighted — the block is passed through `bmass->solve`³⁰ — so r carries the same Jacobian weight as a `div`’s c , not the bare physical value. And it is added *after* the chain multiplication, so it acts directly on f .

Prefer sealing for a homogeneous no-flux wall: it needs no hand evaluation of c , stays exact when c varies in space or time (a literal r is stale the moment the coefficient is recomputed), cannot get the sign wrong, and composes — realising (10) automatically where a hand-written Robin would need $\sum_{k \in S} c_k(b)$ correct. A *genuine* physical Robin wall remains available: seal the `div` and add a Robin with the physical (mass-weighted) r , so the rule reads $\sum_{k \in S} F_k(b) + r f(b) = 0$ with r simply one more member of the sealed set.

²⁷`asgard_coefficients_mats.hpp`:390–394.

²⁸`FENRIS/ICRF_2D/src/ICRF_2D.cpp`:610–614.

²⁹`asgard_pde.hpp`:694–704.

³⁰`asgard_term_build.cpp`:1105–1120; likewise the standalone `term_robin` path, ...:861–864, 870–875.

7 Mass, Jacobians, and multiple dimensions

`set_mass` sets a positive, time-independent volume weight per dimension. The mass matrix is a genuine Galerkin object $M_{rc} = \int \varphi_r m \varphi_c$, Cholesky-factored once and then **inverted into every link of every term**³¹ — so a single term stores $M^{-1}A$ and a chain stores $(M^{-1}A_0)(M^{-1}A_1)\dots$. For one `term_div(c)` the equation actually solved is

$$m(x) \partial_t f = -\partial_x(cf), \quad \text{i.e.} \quad \partial_t(mf) = -\partial_x(cf). \quad (14)$$

The conserved quantity is $\int mf dx$ and its flux is cf , which is why the coefficient handed to `term_div` is effectively mass-weighted: to represent a physical flux Γ in a metric with weight m , pass $c = m\Gamma/f$. (8) is the second-order version of the same statement. The mass multiplies rows on the left and leaves the boundary-flux structure untouched, so everything in §§2–5 holds verbatim with $\int f$ replaced by $\int mf$.

A live trap in the production source. ICRF_1D’s lambdas are *named* `eta_v2` and `zeta_v2` but *compute* $-v^2\eta$ and $-v^2\zeta$.³² The minus signs are essential — they are what makes $c_d c_g = -m^2 D$ come out right — and the names hide them. Read the body, not the name.

Adaptivity. Time-independent terms are assembled *once at max level* and never rebuilt,³³ so the operator the solver sees is the Galerkin *restriction* of a fixed matrix. [INFERENCE]: this is why sealing survives refinement exactly — the level-0 constant is always in the sparse grid, so the row expressing global mass balance is always present, and a block absent at max level is absent in every restriction. Time-dependent terms are rebuilt at the current level, so their $1/h$ and wall blocks change; the wall location and the seal structure do not.

7.1 term_md

A separable `term_md` holds one `term_1d` per dimension, each chain-collapsed and applied as a Kronecker sweep, so the operator is $\bigotimes_d A_d$.³⁴

Sealing a 1-D factor seals the whole term, because the wall integral factorises as (1-D wall block) $\otimes$ (mass in the other dimensions) and zeroing the 1-D block kills the tensor product. [INFERENCE] (structural, from the tensor form, not separately measured): a flag on the `div` of dimension d therefore controls dimension d ’s wall and nothing else.

Flux dimensions are not unique. A `term_md` carries a flux in one dimension *per term_1d factor*; `flux_dim()` returns the *first* it finds, and nothing enforces uniqueness.³⁵ The quasilinear operators in both 2-D solvers are `mode::chain term_mds` carrying fluxes in *two* dimensions at once, e.g. `term_md({div_x, diffusion, grad_K_xtheta})` with the `div` in x and the `grad` in θ .³⁶ [INFERENCE]: an inhomogeneous `boundary_flux` attached to such a term binds to `flux_dim()`’s first hit, silently — a trap for chain-mode terms; read from source, not tested.

Scope of (9). `mode::chain` links are *not* multiplied at build time; they are applied sequentially at run time through a workspace,³⁷ so the intermediate vector *is* materialised and each link’s wall blocks act on a real field. (9) is derived for `term_1d` chains. [UNRESOLVED]: whether it transfers verbatim to `term_md` chains was not established. What is observable is that both production QL operators use the Neumann parity — `div bothsides` with bare `term_grad` links at the default `none` — and that this is the configuration which fixed the measured +63% particle manufacture.

Geometric vanishing. Combining §3’s endpoint sampling with a pitch-angle Jacobian: ICRF_2D’s pitch chain is `term_div(-I(th)*sin(th), ..., bc::bothsides)` with `term_grad(L(th)*sin(th))` on $\theta \in [0, \pi]$.³⁸ The `div` coefficient vanishes at *both* walls, so the block `none` would write is identically zero: `none` and `bothsides` assemble the same matrix and the seal is a no-op. Likewise v^2 at $v = 0$. The wall that matters is the one where the Jacobian does not vanish — and when you add a θ -direction term whose coefficient does *not* carry the $\sin \theta$ factor, its wall block is suddenly live and the sum rule at $\theta = 0, \pi$ acquires a real member.

³¹`asgard_coefficients_mats.hpp:454–492; asgard_term_build.cpp:870–875; asgard_block_matrix.cpp:283+`. An identity dimension gets no matrix and therefore no M^{-1} , `asgard_term_build.cpp:683–687`.

³²`FENRIS/ICRF_1D/src/ICRF_1D.cpp:261–263, 277–279`.

³³`asgard_term_build.cpp:645–651, 716`.

³⁴`asgard_pde.hpp:1081–1096; asgard_term_build.cpp:711–716; asgard_term_manager.cpp:67–68; term_identity` dimensions are skipped, `asgard_kronmult.hpp:41–46`.

³⁵`asgard_pde.hpp:694–712, 1264–1289`; the doxygen claims only one is allowed, but no code checks.

³⁶`FENRIS/LHCD_2D/src/LHCD_2D.cpp:336–339; FENRIS/ICRF_2D/src/ICRF_2D.cpp:739–742`.

³⁷`asgard_term_manager.cpp:103–118`.

³⁸`FENRIS/ICRF_2D/src/ICRF_2D.cpp:510–513, 628–650`.

8 Design recipe

1. **Put your PDE in the convention:** $m\partial_t f = -\partial_x \sum_k F_k$. If it reads $m\partial_t f = +\partial_x \sum_k G_k$, then $F_k = -G_k$.
2. **Read off the coefficients.** An advective channel $F = cf$ is one `term_div(c)`. A diffusive channel $F = -mD\partial_x f$ is a chain with $c_d c_g = -m^2 D$; the split is free but the signs must be *opposite*. Leave `flux_type` alone — `upwind` is the default on both links and the constructor `flip` handles the rest.
3. **Decide the physical wall law at each wall separately:** total flux zero (reflecting), f prescribed (absorbing), $\alpha f + \beta F = 0$ (partially absorbing), or nothing (outflow). “Total” means the sum over *all* terms; you do not get to specify each term’s flux.
4. **Choose the sealed set S per wall** so $\sum_{k \in S} F_k = 0$ is that law, and have every member of S name that wall on its `div`.
5. **Complete each chain per wall:** exactly one of its links names each wall ((9)). If the term is in S that is the `div`, so the `grad` must *not* name that wall.
6. **Audit (§8.3).**

8.1 Worked example, end to end

$\partial_t f = \partial_v (A(v)f + D(v)\partial_v f)$ on $[0, v_{\max}]$, $D > 0$, no Jacobian ($m = 1$), **perfectly reflecting at both ends**. *Step 1:* $\partial_t f = -\partial_v (F_{\text{drag}} + F_{\text{diff}})$ with $F_{\text{drag}} = -Af$, $F_{\text{diff}} = -D\partial_v f$. *Step 2:* $c_{\text{drag}} = -A(v)$; for the diffusion $c_d c_g = -D$, so take $c_d = -D(v)$, $c_g = +1$. *Steps 3–5:* reflecting means total flux zero at both walls, so $S = \{\text{drag}, \text{diffusion}\}$ at both — both `divs` name $\{L, R\}$ and the `grad` names neither.

```
using bc = asgard::boundary_type; // as in the FENRIS solvers
auto c_drag = [](std::vector<P> const &v, std::vector<P> &c) {
    for (std::size_t i = 0; i < v.size(); ++i) c[i] = -A(v[i]); // c = -A
};
auto c_div = [](std::vector<P> const &v, std::vector<P> &c) {
    for (std::size_t i = 0; i < v.size(); ++i) c[i] = -D(v[i]); // c_d = -D
};
pde += term_1d({term_div(c_drag, bc::bothsides)}); // drag, sealed
pde += term_1d({term_div(c_div, bc::bothsides), // c_d * c_g = -D
                term_grad(P{1})}); // c_g = +1, flag = none
```

Realised wall law, from (10) at each wall: $-Af - Df' = 0$, i.e. $f'/f = -A/D$ — the emergent Robin, not $f' = 0$. Mass is conserved structurally, $\dot{N} = 0$ to round-off at every level.

The one-line diff for absorbing at $v_{\max}$, reflecting at $v = 0$. The walls are independent (§4.3). At L the flux stays sealed; at R we want $f = 0$ with the flux free. So the `divs` name $\{L\}$ and the `grad` names $\{R\}$:

```
pde += term_1d({term_div(c_drag, bc::left)}); // sealed at L, free at R
pde += term_1d({term_div(c_div, bc::left), // pins f = 0 at R
                term_grad(P{1}, bc::right)});
```

Check (9): $S_d = \{L\}$, $S_g = \{R\}$, complementary at both walls. Left: reflecting, $f'/f = -A/D$, no leak. Right: $f \rightarrow 0$, flux free, particles absorbed.

Is the free drag at the absorbing wall a problem? Audit item 5 flags unsealed advective terms whose characteristic points inward, and a Fokker–Planck drag at $v_{\max}$ does point inward. It is harmless here, but only asymptotically, and the reason is worth stating: the `grad`’s pin drives the *interior trace* $f^{\text{int}}(v_{\max})$ to zero, and the free advective block is $c f^{\text{int}}$, so the destabilising flux goes to zero with it. Measured for the pure-diffusion Dirichlet pair, the wall value converges to zero at $O(h^3)$ ($-5.10\text{e-}07 \rightarrow -6.37\text{e-}08 \rightarrow -7.97\text{e-}09$ over levels 5–7). [INFERENCE] that this suffices to tame a co-located free advective term: the measured run has no advective channel. **Audit item 5 applies only at walls where the value is not otherwise pinned.**

8.2 Worked cases

physical wall law	sealed set S	flags	realised condition
Reflecting	<i>all</i> flux-carrying terms	bothsides on every div ; grads left none	$\sum_k F_k = 0$; with drag this is $f'/f = -A/D$, <i>not</i> $f' = 0$
Dirichlet $f = 0$	$\emptyset$	{ div none , grad bothsides }; advective terms none	$f(\text{wall}) = 0$, correct Dirichlet spectrum
Mixed (reflect L , absorb R)	all terms, at L only	{ div left , grad right }	§8.1
Prescribed in-flow flux $F = g$	the term carrying the inflow	bc::left <i>plus</i> left_boundary_flux(g)	$\hat{F}_a = g$. <i>Not</i> $f(a) = g$, unless the term is a bare advective div
Inhomogeneous Dirichlet $f = g$	$\emptyset$	{ div none , grad bothsides }, boundary flux attached at the grad via chain_level	[INFERENCE]: mechanism read from source (§3.3), combination untested
Partially absorbing $\alpha f + \beta F = 0$	all flux terms, plus the Robin	bothsides on every div , then set_right_robin(r) , $r \geq 0$, mass-weighted, scalar, chain only	$\sum_{k \in S} F_k + r f = 0$
Outflow	$\emptyset$	none on the advective div ; keep any chain legal	$\hat{F} = c f^{\text{int}}$; <i>only</i> valid if c points out of the domain
Pure Neumann $f' = 0$ (rarely wanted)	the diffusion terms only	bothsides on the diffusion divs , none on the drag	$f' = 0$, and the drag then leaks — usually a bug

8.3 The audit

One row per term, one block per wall. Example at a right wall b with a drag whose $c(b) = -4$ (inward, the FENRIS-typical case):

term k	flux F_k	$F_k(b)$ (direction)	flag	sealed?	contributes to $\dot{N}$
drag	$c_{\text{drag}} f$	$-4f(b)$ (inward)	bothsides	yes	—
collisional diffusion	$-D_1 f'$	$-D_1 f'(b)$	bothsides	yes	—
QL diffusion	$-D_2 f'$	$-D_2 f'(b)$	none	no	$+D_2 f'(b)$
leak rate $\dot{N} _b = -\sum_{\text{unsealed}} F_k(b) \rightarrow$					$+D_2 f'(b)$

1. **The unsealed column gives the leak rate**, via (11): $\dot{N} = \sum_{k \notin S} [F_k(a) - F_k(b)]$. For an impermeable wall it must be empty. Had the drag row been unsealed above it would contribute $-F_{\text{drag}}(b) = +4f(b) > 0$: mass *manufactured*, the ICRF/LHCD pathology.
2. **The sealed column gives your wall law**, and it is the only one you get. Solve $\sum_{k \in S} F_k = 0$ and check it is the physics you wanted; if $\sum_{k \in S} D_k = 0$ you have accidentally imposed Dirichlet.
3. **Per chain, per wall: exactly one link names that wall** ((9)).
4. **Across terms, per wall: either S is non-empty or some grad names that wall — never both.** A **grad bothsides** imposes $f = 0$ there for the whole problem; combining it with a non-empty sealed set puts two conditions on one scalar (§4.3(iii), blow-up).
5. **Direction.** For every unsealed advective term at a wall where the value is not otherwise pinned, does the characteristic leave the domain? If it points inward, the free flux is a positive feedback on the wall value.
6. **Signs.** Per chain, c_d and c_g opposite (§4.2); per Robin, $r \geq 0$ at a right wall, $r \leq 0$ at a left one.
7. **Endpoint value.** If the coefficient vanishes there, the flag is inert and the row is a no-op.

Where the recipe does not reach, named because the failure is silent: **(a)** a wall law whose coefficient varies *along* a wall — seal/free is a per-term, per-dimension binary and r is a scalar, so the only mechanism admitting dependence on another coordinate is **boundary_flux** (§3.3); **(b)** a genuinely non-separable wall flux, which must be decomposed into separable **term_mds**; **(c)** cross-derivative fluxes (Γ_x containing $\partial_\theta f$), which are the chain-mode **term_mds** of §7.1; **(d)** translating a reflection coefficient into $\alpha f + \beta F = 0$, since the DG wall exposes only $f(\text{wall})$

and one *net* flux, never separated incoming and outgoing halves. For (c) there is one production pattern worth copying, recorded and measured in ICRF_2D: the QL flux components look coupled, but because the QL tensor is rank-1, *zeroing both components* forces $\mathbf{v} \cdot \nabla f = 0$ and the entire QL flux drops out, leaving a purely collisional wall law with no θ dependence. Sealing both directions can be simpler than analysing either.³⁹

9 Pitfalls

P1. Nothing validates flag combinations. No check anywhere that a chain satisfies (9). The over-determined pair compiles, builds, runs and blows up; the under-determined pair compiles, builds, runs and quietly produces wrong physics.

P2. Mass conservation does not detect the over-determined pair. With `div bothsides` the flux block is absent, so there is *no mechanism* for mass to leave, whatever the `grad` does. Measured at level 7: mass error 6.3×10^{-9} absolute — 1.4×10^{-8} relative to the conserved mass 0.443, 4×10^{-15} relative to $\max|f|$ — while f diverges to 1.4×10^6 as h^{-3} . The absolute error *grows* 10–20× per level ($2.9\text{e-}11 \rightarrow 5.1\text{e-}10 \rightarrow 6.3\text{e-}09$), which sharpens the point rather than softening it. **Add a $\max|f|$ assertion.**

P3. A one-sided bad flag corrupts the whole solution, not just its wall: the good wall reads 0.201417 against the clean Neumann 0.237950, 15% off.

P4. `flux_type::none` is silently central. The enumerator has no initialiser, so it takes `downwind + 1 = 0` — indistinguishable from `central`. Hence $s = 0$, and the dispatch calling `gen_no_flux_cmat` has **no return**, so execution falls through and the full central-flux matrix is assembled on top.⁴⁰ **[INFERENCE]** (strong): the no-flux path is dead, and `flux_type::none` gives a central flux *with* full edge terms, including a central-flux boundary block. Waste, not corruption (the stale data is zeroed), and no FENRIS solver uses it.

P5. Vanishing Jacobians make flags inert, so a passing test at such a wall says nothing about your flags (§3).

P6. c_d and c_g must have opposite signs, per chain — equivalently each chain’s physical diffusivity $D_k = -c_d c_g / m^2$ must be ≥ 0 *separately*. You cannot split a diffusion operator into a positive and a negative chain even when the net diffusivity is positive: each chain’s upwind split $c \pm |c|$ is built from *its own* coefficient and the wrong sign picks the anti-diffusive pairing (§4.2). Measured with $D_1 = +1$ fixed: $D_2 = +0.3$ behaves perfectly (1.2×10^{-4} at level 6, the expected $O(h^2)$); $D_2 = -0.05$ — net diffusivity still $+0.95$ — blows up by 13 orders of magnitude in $t = 0.2$; $D_2 = -0.3$ reaches $\max|f| = 4.3 \times 10^{76}$. The growth is **interior** ($x \approx 1.1$), not at a wall. State this loudly, because the sum rule *invites* the mistake: drags of opposite sign cancel across terms perfectly well (§5); diffusions do not.

P7. `div(bothsides)` plus a penalty is over-determined ([INFERENCE], §3.2). **P8. Free flux at an inflow wall** is a Robin of the destabilising sign and does *not* improve with refinement — unless the wall value is pinned by something else (§8.1). **P9. The Robin API is more restrictive than it looks:** scalar, compile-time, chain-only, mass-weighted. **P10. `flux.dim()` returns the first flux dimension, not the only one** (§7.1).

P11. Measure wall slopes off the DG polynomial, not off a plot grid. Finite-differencing an interpolated grid over an interval comparable to h produced apparent 21% errors in a quantity converged to four digits (see the corrections table). For degree 2 the solution is a quadratic on each cell, so fitting a quadratic to 10 interior points of the last cell recovers f and f' at the wall to round-off.

10 Verification

Every row compares a measurement against an *independently known analytic* value. Protocol, commands, per-test tables, reproduction instructions and the recommended regression checks are in `03_experiments.md`; only the summary is reproduced here.⁴¹ Toys are written as $\partial_t f = \partial_x G$, so in this document’s convention $F = -G$ (e.g. $c_{\text{drag}} = -2x$); dN/dt is convention-independent and is tabulated directly.

³⁹FENRIS/ICRF_2D/src/ICRF_2D.cpp:585–600.

⁴⁰asgard_pde.hpp:65–75; asgard_coefficients_mats.hpp:123–128, 155.

⁴¹One harness gotcha worth carrying over: a deck file is read *only* when passed with `-if/-infile`. `./solver deck.txt -of out.h5` runs with every key at its default and no error message; deck values override CLI options appearing *before* `-if`.

claim	predicted	measured	error
<i>Neumann pair</i> {div bothsides, grad default}			
eigenvalue, 4 combinations of (L, k, D)	$D(k\pi/L)^2$	agrees	10^{-9} – 10^{-7}
mass conservation, levels 4–8	exact	$8.9\text{e-}16 \rightarrow 2.7\text{e-}15$	structural
wall derivative under refinement	$\rightarrow 0$ at $O(h^{k+1})$	$9.7\text{e-}08 \rightarrow 2.4\text{e-}11$	rate exactly 3
<i>Dirichlet pair</i> {div none, grad bothsides}			
ground-mode decay rate, $L=2, D=1$	$(\pi/2)^2 = 2.467401$	2.467401	$5.1\text{e-}09$
2nd, 3rd eigenvalues; $L=3; D=0.4$	$\pi^2, (3\pi/2)^2, (\pi/3)^2,$ $0.4(\pi/2)^2$	all agree	$\leq 4.1\text{e-}07$
profile relaxes to $\sin(\pi x/2)$; wall value $\rightarrow 0$	shape, then $O(h^{k+1})$	L_2 dev $3.1\text{e-}07$; $5.1\text{e-}07$ $\rightarrow 8.0\text{e-}09$	rate 3
absolute decayed mass vs Fourier series	0.0410189977	0.0410189972	$1.3\text{e-}08$
<i>Sum rule</i> , $\partial_t f = \partial_x(2xf + D_1 f' + D_2 f')$, all 8 sealings, level 8			
$S=\{A, D_1, D_2\}$: $f'/f = -A_S/D_S$	-2.66667	-2.66664	$9.3\text{e-}06$
$S=\{A, D_1\}$; $S=\{A, D_2\}$	-4.00000 ; -8.00000	-3.99996 ; -7.99990	$1.1\text{e-}05$; $1.2\text{e-}05$
$S=\{A\}$, $D_S=0 \Rightarrow$ Dirichlet	$f(2) = 0$	$1.67\text{e-}09$	$3.4\text{e-}10$ rel
S purely diffusive $\Rightarrow$ Neumann (3 cases)	$f' = 0$	$1.4\text{e-}07$	$1.4\text{e-}07$
$\dot{N}$ = open-channel flux, all 8 sealings	—	agrees; fully open gives $+4298$	$\leq 1.7\text{e-}04$
one f' channel sealed silences all	identical $\dot{N}$	$9.01295/9.01295/9.01294$	6 figures
wall slope under refinement, levels 4–8	$O(h^2)$	rates 2.00 – 2.02 ; steep- est case (-8) $3.8\text{e-}3 \rightarrow$ $1.2\text{e-}5$	$2.5\text{e-}03 \rightarrow 9.3\text{e-}06$
<i>Sum rule is signed and local</i>			
sealed set with $A_S = -1$ (sign flip)	$f'/f = +0.66667$	$+0.66667$	$2.8\text{e-}07$
two nonzero drags cancelling, $A_S=0$	$f' = 0$ (Neumann)	$3.0\text{e-}13$	—
$D_2(x)=0.25x$ (same wall value, different interior); $D_2(x)=0.5+0.75x$ (D_S doubled at wall)	-2.66667 ; -1.33333	-2.66663 ; -1.33333	$1.3\text{e-}05$; $2.0\text{e-}06$
<i>Robin</i> , $\partial_t f = \partial_x(2xf + f')$ on $[0, 2]$, level 7, $t=4$			
Robin($r=4$) vs sealing the drag	identical matrices	max abs diff $2.1\text{e-}13$	—
sealed profile vs analytic Ce^{-x^2} ; mass	0.8820813908	L_2 rel $7.2\text{e-}08$; all 10 digits	—
drag free with <i>no</i> Robin	0.88208	1333.97 ($1512\times$)	fails, as designed
<i>All six chain pairings</i> , $\partial_t f = \partial_x f'$, exact IC mass 0.443096843048			
Neumann: mass at $t=1$, levels 5–7	0.4430968430	0.4430968430	$\leq 5.0\text{e-}16$
over-determined: max $ f $ at level 7	—	$1.42\text{e+}06$, grows as h^{-3}	mass still exact
one-sided grad right : which wall	diverges at R	$f(2)=1.42\text{e+}06$, $f(0)$ bounded	—
under-determined: $f(0)$; $f(2)$	—	$\rightarrow 0.192$; $\rightarrow 0^-$ at $O(h)$	19% off Neumann
Dirichlet: mass at levels 5–7	Fourier 0.0410189977	0.0410189972	$1.3\text{e-}08$
<i>Negative-coefficient control</i> (P6), level 6, $t=0.2$, $D_1 = +1$			
$D_2 = +0.30$	$f'/f = -3.076923$	-3.076568	$1.2\text{e-}04$
$D_2 = -0.05$ (net $+0.95$); $D_2 = -0.30$	-4.210526 ; -5.714286	max $ f = 2.0\text{e+}13$; $4.3\text{e+}76$	diverge, interior

One measured asymmetry that is *not* explained. In the Neumann refinement study $|f'(0)|$ is consistently $2.5\times |f'(2)|$, level independently, both converging at the same $O(h^3)$ rate. An asymmetry is expected — the alternating flux takes $\hat{f}$ and $\hat{q}$ from opposite sides, so the two walls are not equivalent — but the *value* 2.5 is not derived anywhere and its dependence on degree is untested. [UNRESOLVED]: treat the measured facts (level-independent ratio, equal rates) as the claim, not the attribution.

Corrections to earlier claims. Three claims from an earlier session do not reproduce as stated. None overturns the mechanism or the sum rule, but all three would mislead someone debugging a real solver.

earlier claim	status	correct statement
sum-rule slopes measured −2.60, −3.69, −6.3 against predictions −2.67, −4, −8; the last “under-resolved”	measurement artifact	at the <i>same</i> run parameters, reading f'/f off the DG polynomial gives −2.66627, −3.99931, −7.99840 (rel. err $1.5\text{--}2.0 \times 10^{-4}$). There is no boundary layer needing extra refinement.
the over-determined pair has “mass error 8×10^2 ”	wrong	mass is conserved (1.4×10^{-8} relative to the conserved mass). The pathology is <i>pointwise divergence</i> , <i>not mass loss</i> , so conservation diagnostics do not catch it.
with a one-sided grad flag “the other wall stays exactly Neu- mann”	too strong	the <i>divergence</i> is local, but the good wall reads 0.201417 against 0.237950: 15% off. The whole solution is wrong.